

# INTRO TO SYNTHESIS

A crash course in subtractive synthesis, on the Sequential Fourm





# TODAY

Club Control, Bucharest

## SINTEZAUR

Bucharest's community for people who love synthesis. Your guide today: Iulian.

## ZEEDO

Our retail partner, and the reason there are a dozen synths in this room.

## SEQUENTIAL FOURM

Today's instrument: a 4-voice analog synth that feels pressure in every key.

SINTEZAUR × ZEEDO · POWERED BY SEQUENTIAL

WHERE ON FOURM

# TRANSPOSE DOWN + PROGRAM

Hold TRANSPOSE DOWN (left of the keys) and press PROGRAM: the synth resets to the Basic Program.



# SIGNAL FLOW

How sound travels through a subtractive synthesizer



# WHAT'S IN A SOUND

A note is a fundamental plus a stack of fainter harmonics. Their mix is the timbre.

BAR HEIGHT = HOW LOUD · LEFT TO RIGHT = HIGHER IN PITCH



## SAW

EVERY HARMONIC

*bright, buzzy*



## SQUARE

ODD HARMONICS ONLY

*hollow, woody*



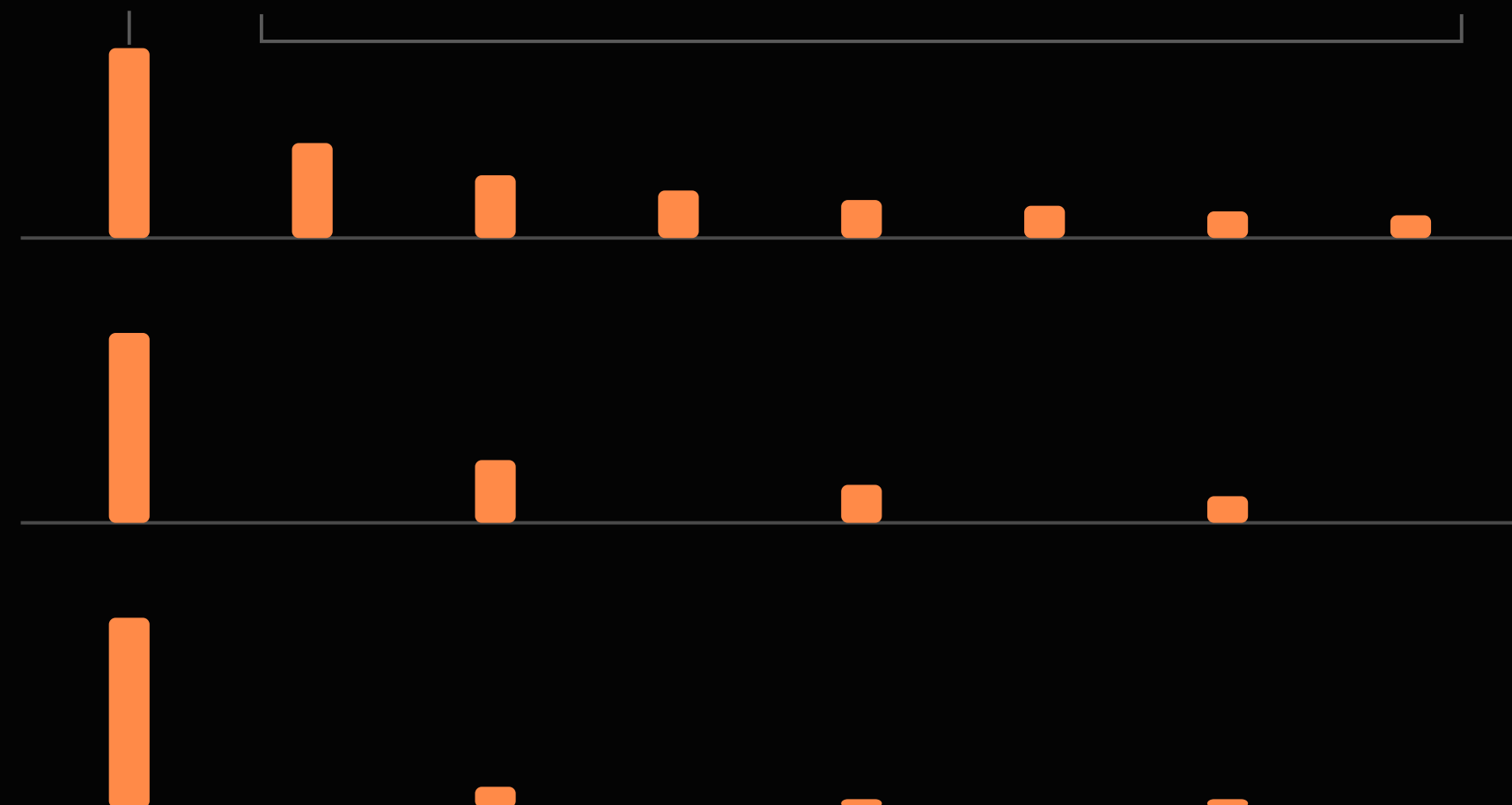
## TRIANGLE

ODD, FADING FAST

*soft, nearly a sine*

FUNDAMENTAL

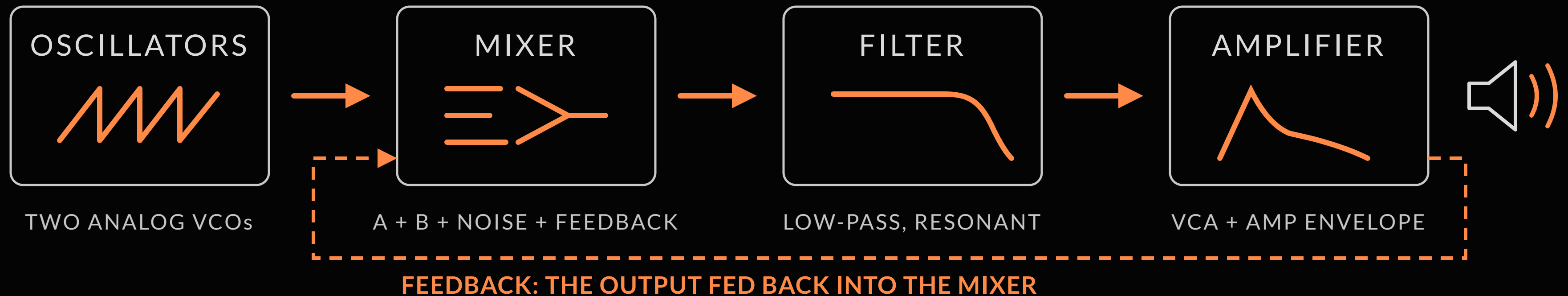
HARMONICS



SQUARE IS PULSE AT 50%. OFF 50%, EVEN HARMONICS APPEAR — THAT'S WHAT PWM MOVES.

# THE VOICE PATH

Oscillators, mixer, filter, amplifier: one road for every sound



# WHERE ON FOURM

The signal path, left to right across the panel



# THE MIXER

Every voice starts here: four sources feed the filter

## OSC A



SAW + PULSE  
THE BRIGHT BACKBONE

## OSC B



TRI + SAW + PULSE  
WEIGHT, DETUNE, LO MODES

## NOISE



WHITE / PINK / VIOLET  
AIR, PERCUSSION, TEXTURE

## FEEDBACK



OUTPUT LOOP  
SOFT-CLIP SATURATION

**TURN UP AT LEAST ONE SOURCE, OR LET RESONANCE SELF-OSCILLATE**

# KEY TERMS

New words from this module

## SIGNAL FLOW

The road a sound takes: oscillators → mixer → filter → amplifier → output.

## VCO · OSCILLATOR

Voltage-controlled oscillator. Generates the raw waveform. Fourm has two.

## MIXER

Sets the level of each source: Osc A, Osc B, noise, feedback.

## HARMONICS

Whole-number multiples of the fundamental, stacked above it and usually fainter; their mix is timbre.

## NOISE

Unpitched sound, in white, pink, or violet flavors.

## FEEDBACK

The output fed back into the mixer, soft-clipped: warmth, then growl.

## VCF · VCA

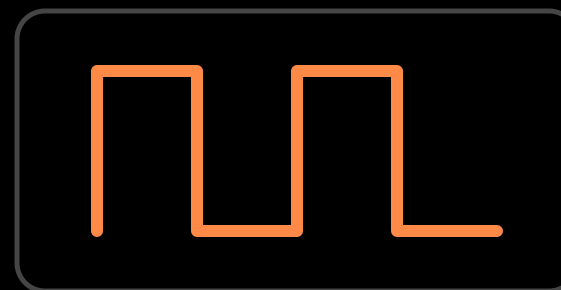
Voltage-controlled filter and amplifier: the tone and loudness stages.

# OSCILLATORS

Where every sound is born



SAW



PULSE



TRI

# THE WAVESHAPES

Two oscillators, five buttons, endless timbres



EVEN + ODD HARMONICS · BRIGHT & BUZZY



ADDS WEIGHT, DETUNE, LO MODES

**PRESS SEVERAL SHAPE BUTTONS AT ONCE: THE WAVES COMBINE**

# OCTAVES & TUNING

Detune B against A and the sound starts to breathe.

-2

-1

0

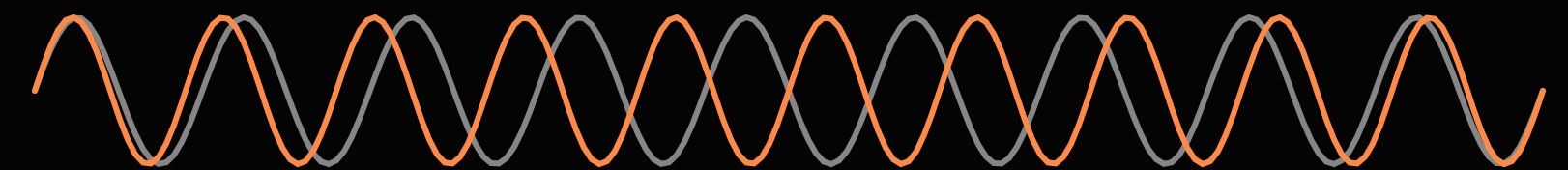
+1

+2

FIVE OCTAVE POSITIONS ON EACH OSCILLATOR

OSC B ONLY: LO1 · LO2, BELOW THE OCTAVES —  
SUB-AUDIO, IT BECOMES A MODULATOR (LEVEL 2)

FREQUENCY · FINE TUNE ·  $\pm 7$  SEMITONES



A AND B, A FEW CENTS APART



TOGETHER THEY BEAT — THE SOUND BREATHES

**TOO FAR AND YOU HEAR TWO NOTES — BRING IT BACK UNTIL IT JUST SHIMMERS**

WHERE ON FOURM

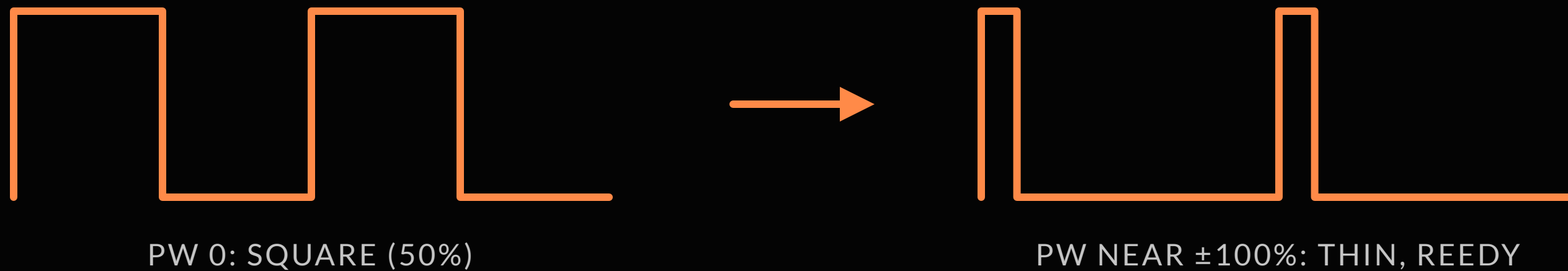
# OSC B › OCTAVE · FREQUENCY

OCTAVE is the encoder. Turn FREQUENCY slowly off centre: slow beats, chorus, two notes — then back.



# PULSE WIDTH & PWM

No PW knob: hold PULSE, turn SELECT. 0 = square. Toward  $\pm 100$ : thin, then silence.



AT FULL  $\pm 100\%$  THE PULSE NARROWS TO SILENCE

**ROUTE LFO OR FILT ENV TO PW: THAT MOVEMENT IS PWM**

WHERE ON FOURM

# OSC A › PULSE|PW + SELECT

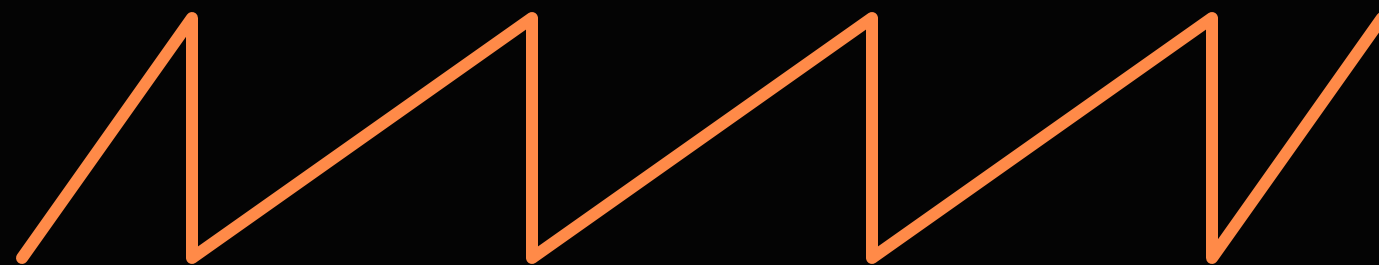
Hold PULSE|PW and turn SELECT: zero is the square, towards the edge the pulse thins to a thread.



# THE SUB TRICK

Osc B, one octave down, triangle

OSC A: SAW · OCTAVE 0



OSC B: TRI · OCTAVE -1



TOGETHER

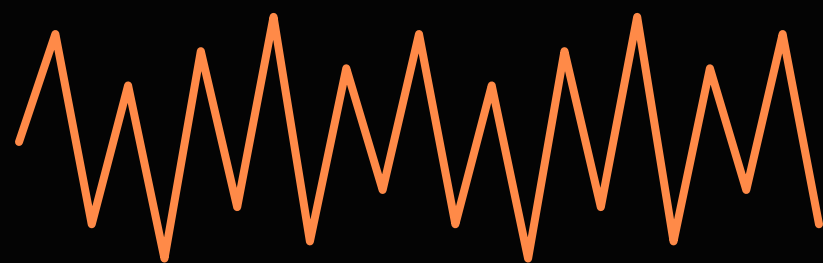


THE TRIANGLE REINFORCES THE FUNDAMENTAL: WEIGHT WITHOUT MUD

# NOISE

Three colors of texture: white, pink, violet

## WHITE



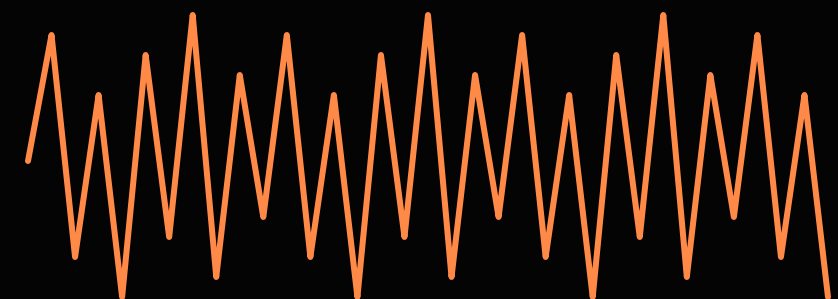
FLAT: HISS, SNARE

## PINK



WARM: WIND, WAVES

## VIOLET



BRIGHT: AIR, SIZZLE

**HOLD NOISE + TURN SELECT FOR LEVEL • TYPE LIVES IN THE PARAM MENU**

# KEY TERMS

New words from this module

## WAVESHAPE

The geometric shape of one cycle: it sets the raw timbre.

## PWM

Pulse width modulation: moving PW with the LFO or an envelope.

## LO1 / LO2

Osc B's low modes: LFO-range pitch, key-tracked (LO1) or free (LO2).

## PULSE WIDTH (PW)

Duty cycle of the pulse wave, -100% to +100%. 0 = square.

## SYNC

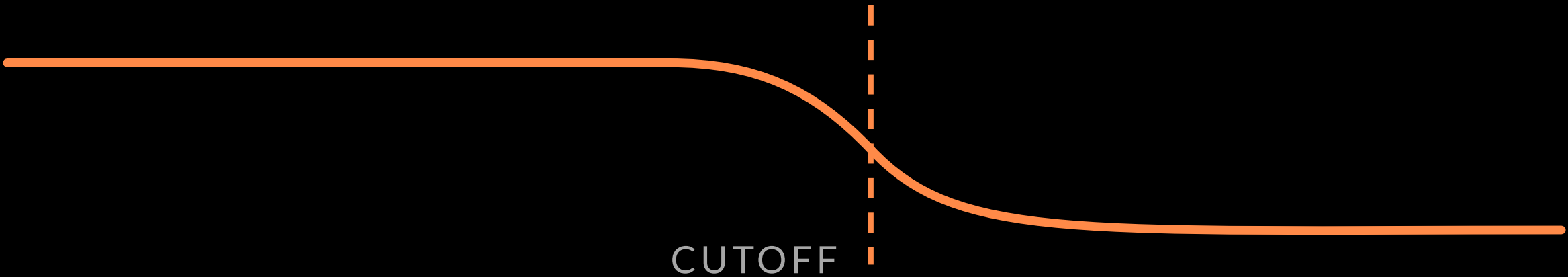
Locks Oscillator A's cycle to Oscillator B for aggressive timbres.

## HARMONICS

Saw has them all; square and triangle only the odd ones — the triangle's fade much faster.

# FILTER

Shaping your tone



FILTER

ENV AMT

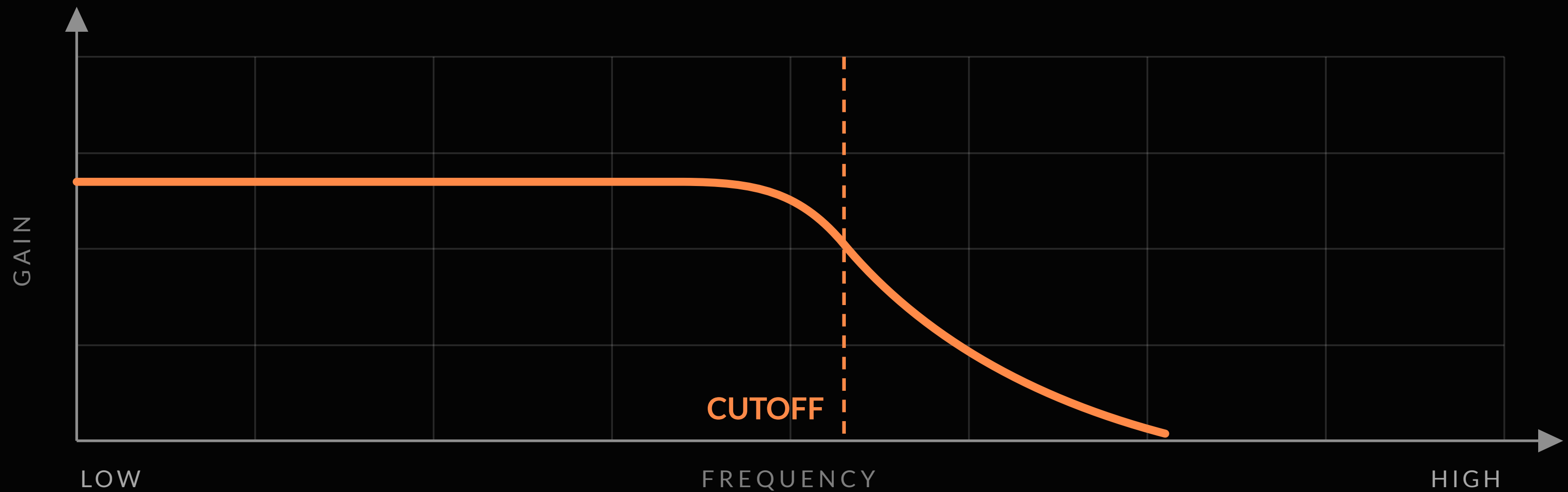
KEY TRACK AMOUNT

CUTOFF

RESONANCE

# LOW-PASS

Below the cutoff passes, above it fades. Close it: darker. Open it: brighter.



WHERE ON FOURM

# FILTER › CUTOFF

Close CUTOFF slowly, then open it again: the harmonics leave and come back one by one.



# RESONANCE

A boost right at the cutoff. Turn it up and the filter sings; at maximum it becomes a sound source of its own.



WHERE ON FOURM

# FILTER › RESONANCE

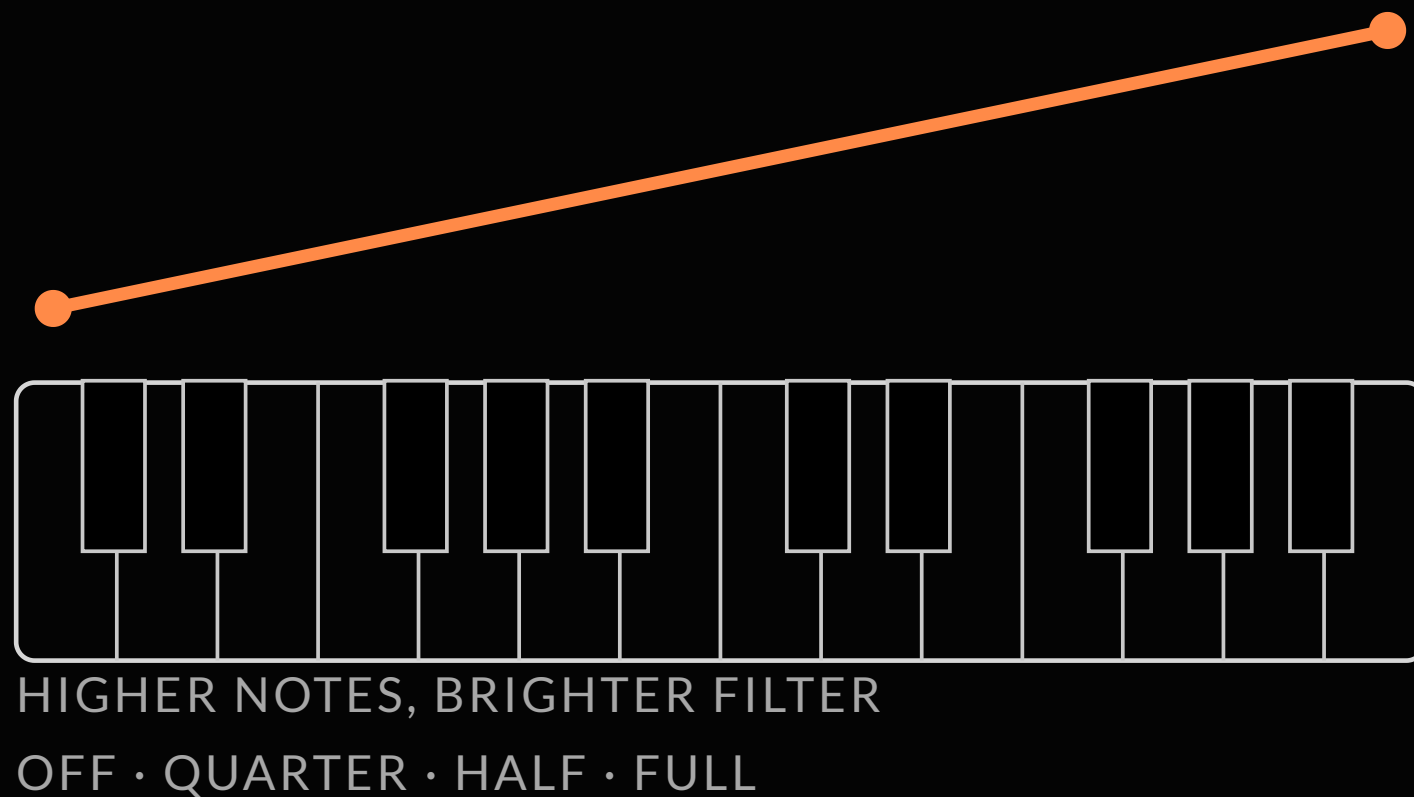
RESONANCE up while you sweep CUTOFF: a peak that whistles along; all the way up, the filter sings on its own.



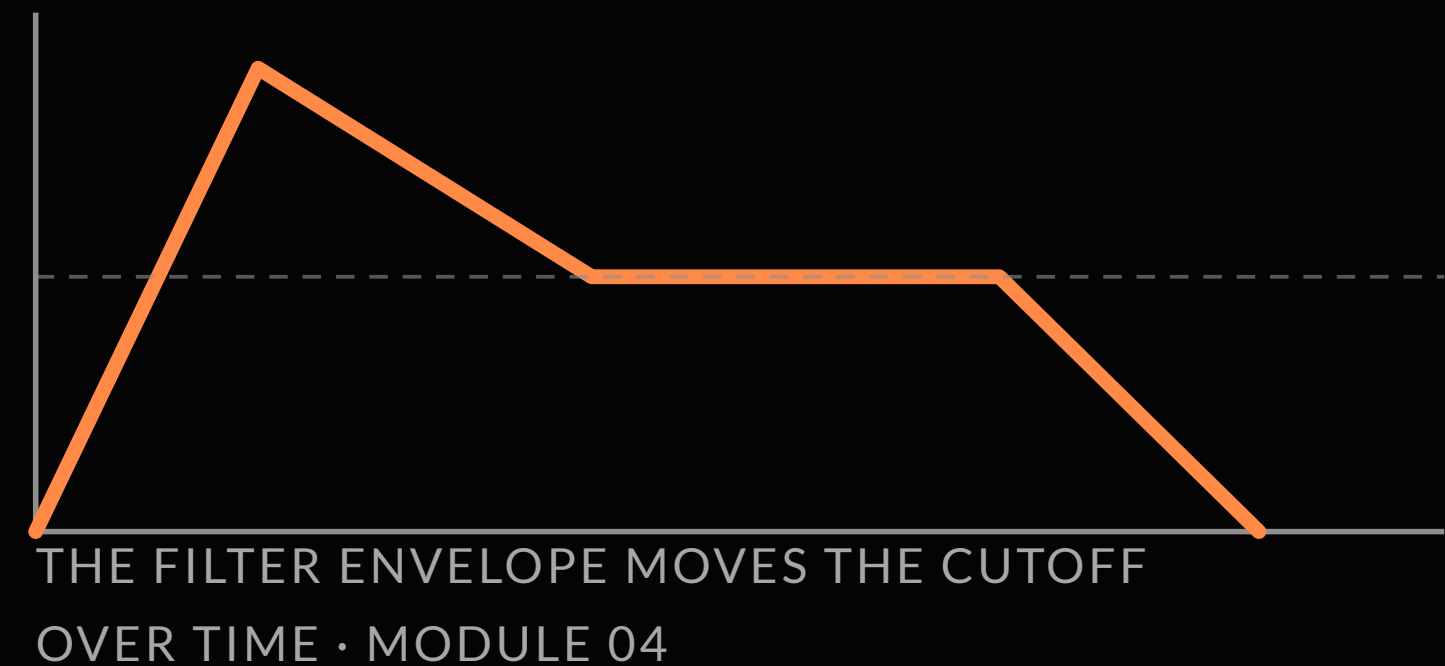
# KEY TRACK & ENV AMOUNT

The keyboard opens the filter; the envelope moves it in time.

## KEY TRACK



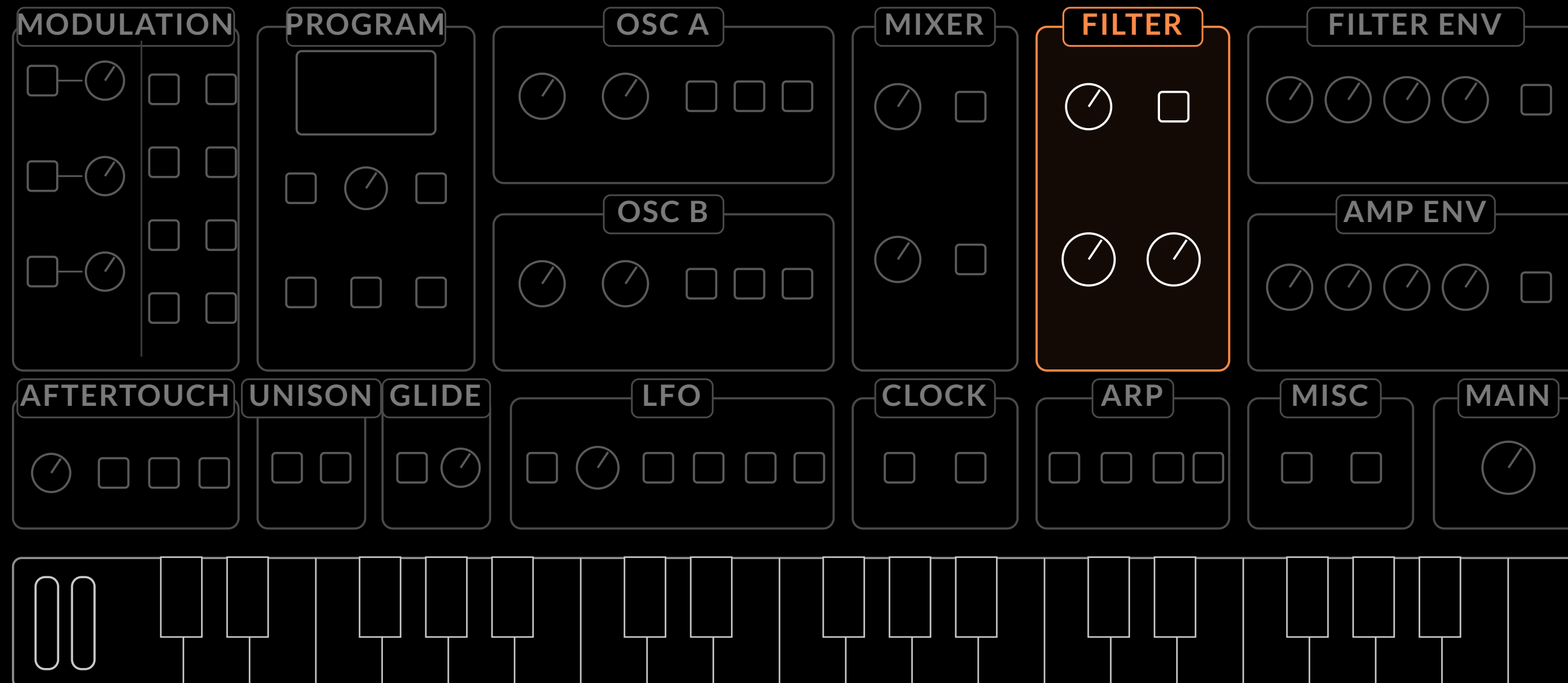
## ENV AMT



WHERE ON FOURM

# FILTER › KEY TRACK · ENV AMT

KEY TRACK on: high notes open the filter more. ENV AMT: how far the envelope pushes the cutoff.



# KEY TERMS

New words from this module

## **CUTOFF**

The hinge: below passes, above fades.

## **SELF-OSCILLATION**

At maximum, the filter becomes a sine source.

## **ENV AMOUNT**

How much the envelope moves the cutoff.

## **RESONANCE**

A boost right at the cutoff.

## **KEY TRACK**

Higher notes, brighter filter.

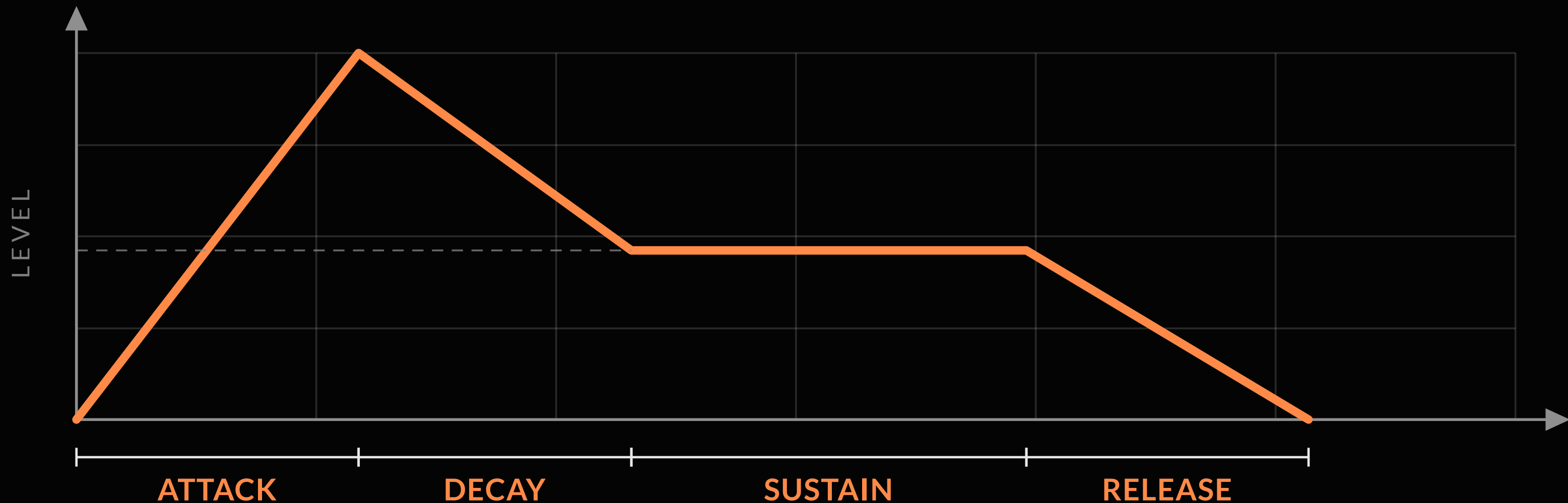
# ENVELOPES

Controlling your sound over time



# ADSR

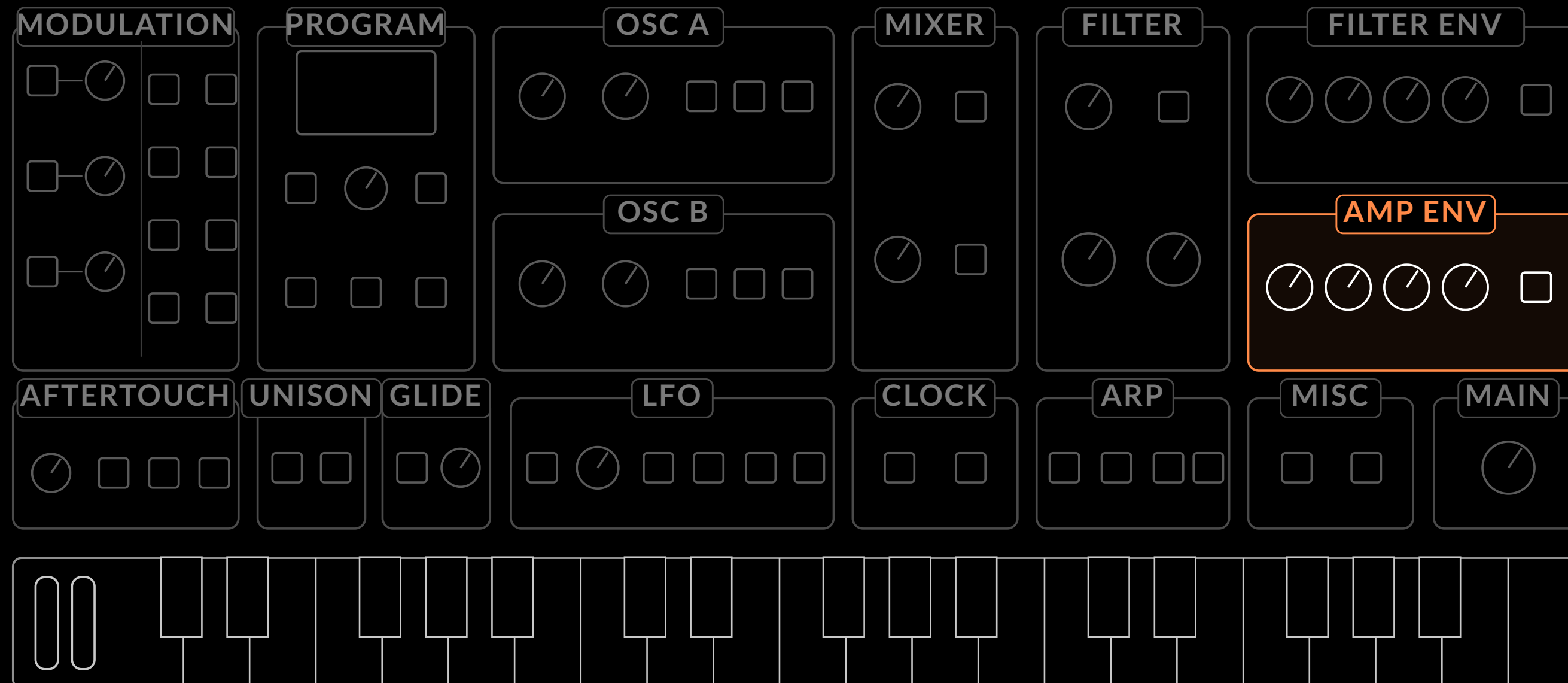
A shape that happens every time you press a key. Sustain is a level, not a time.



WHERE ON FOURM

# AMP ENVELOPE › ATTACK · DECAY · SUSTAIN · RELEASE

Four knobs, one shape. Hold a note and turn ATTACK up, then RELEASE — the note learns to breathe.



# TWO ENVELOPES

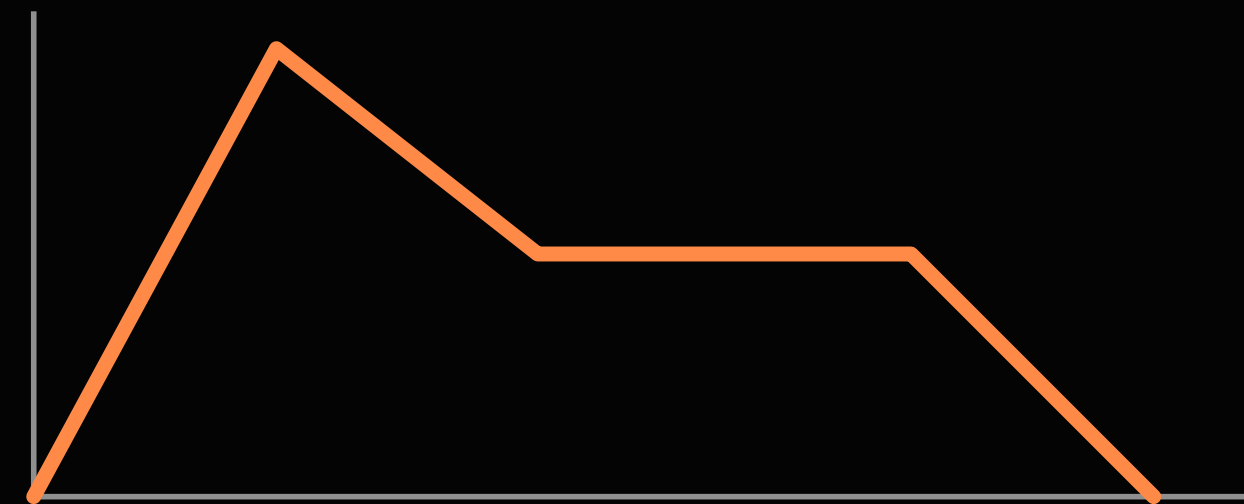
Same shape, two jobs: how loud, how bright.

## AMP ENVELOPE



LOUDNESS  
**HOW LOUD**

## FILTER ENVELOPE

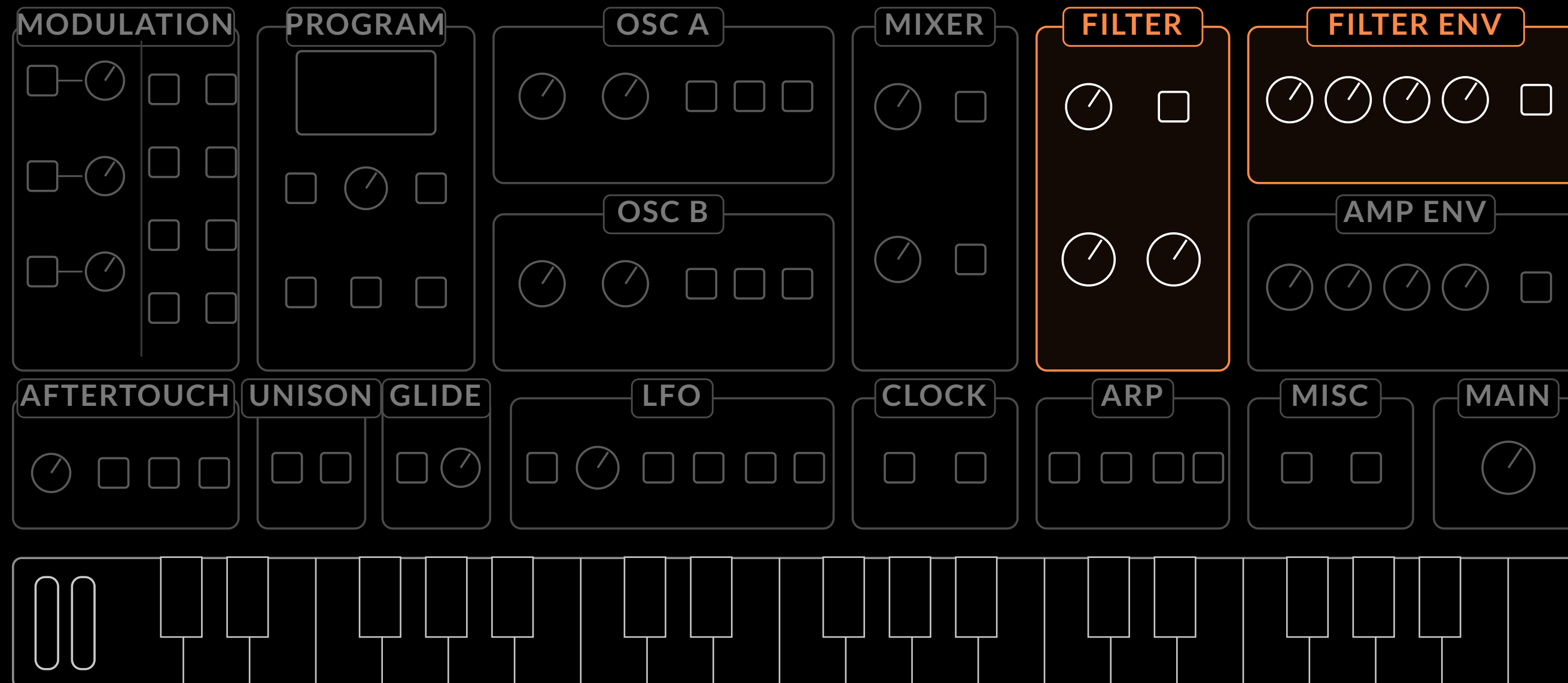


CUTOFF · SCALED BY ENV AMT  
**HOW BRIGHT**

WHERE ON FOURM

# FILTER ENV › DECAY + FILTER › ENV AMT

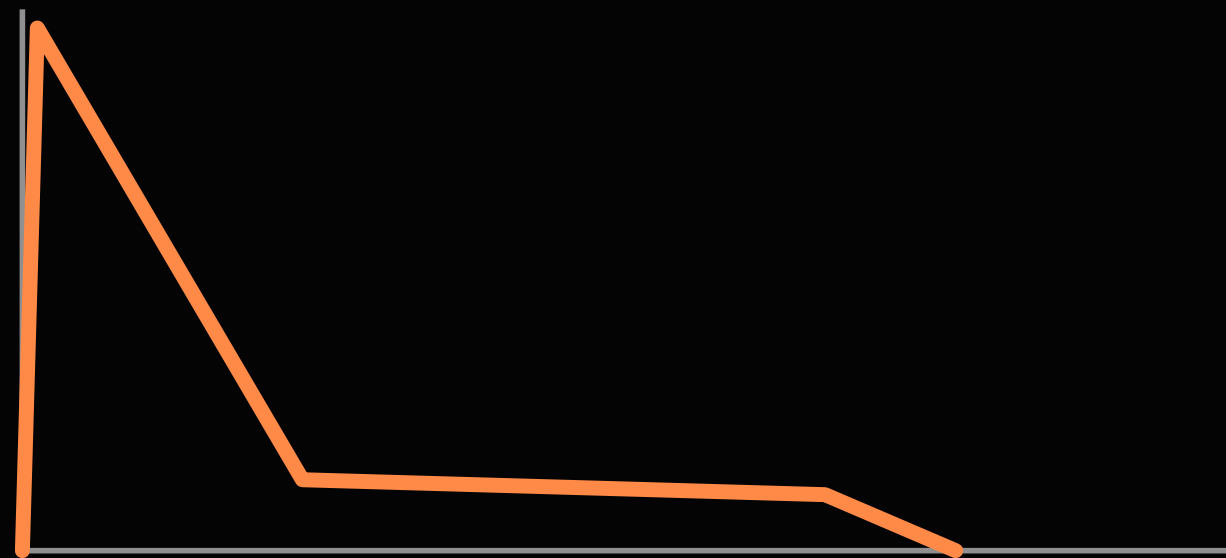
ENV AMT (in the filter) sends the envelope to the cutoff; DECAY sets how fast the brightness falls away.



# BASS vs PAD

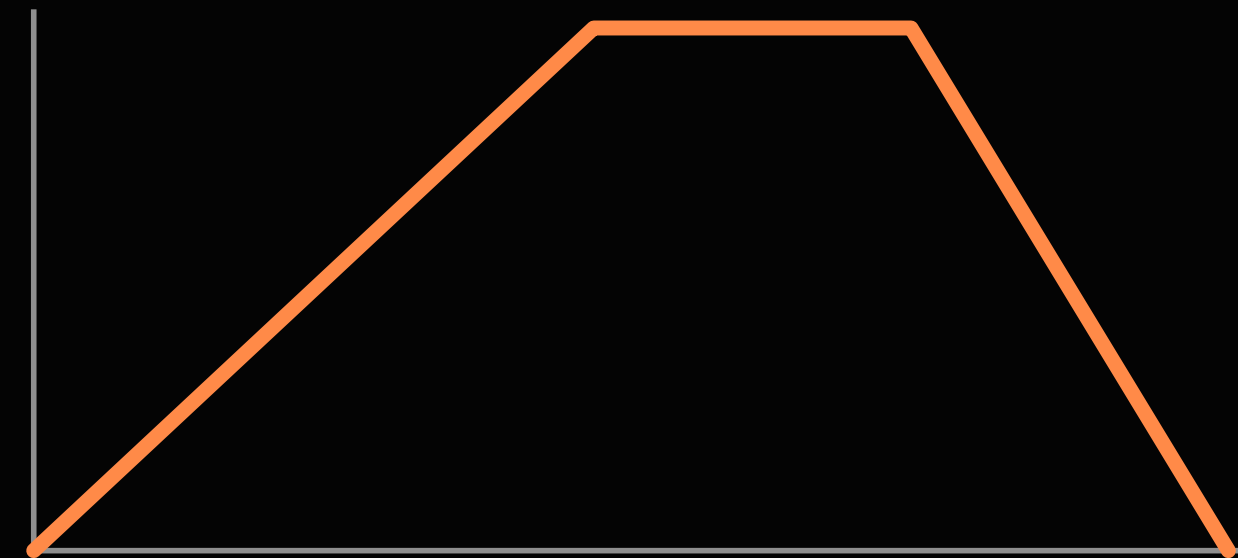
The same oscillators, two instruments.

## BASS



INSTANT ATTACK · SHORT DECAY  
LOW SUSTAIN

## PAD



SLOW ATTACK · NO DECAY  
FULL SUSTAIN · LONG RELEASE

# KEY TERMS

New words from this module

## **ATTACK**

Time from silence to the peak.

## **SUSTAIN**

A level, not a time.

## **ENV AMOUNT**

How much the envelope moves the cutoff.

## **DECAY**

Time from the peak to the sustain level.

## **RELEASE**

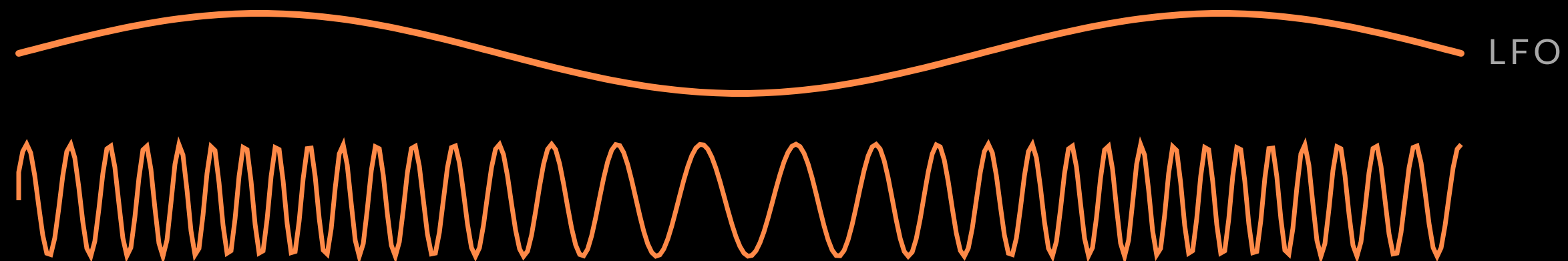
Time to fade after the key is let go.

## **VELOCITY**

How hard you strike scales the envelope.

# LFO & MODULATION

Adding movement



# THE LFO

An oscillator too slow to hear. You hear what it moves.

**LFO**

→ PITCH **VIBRATO**



→ CUTOFF **WAH**



→ AMP **TREMOLO**



WHERE ON FOURM

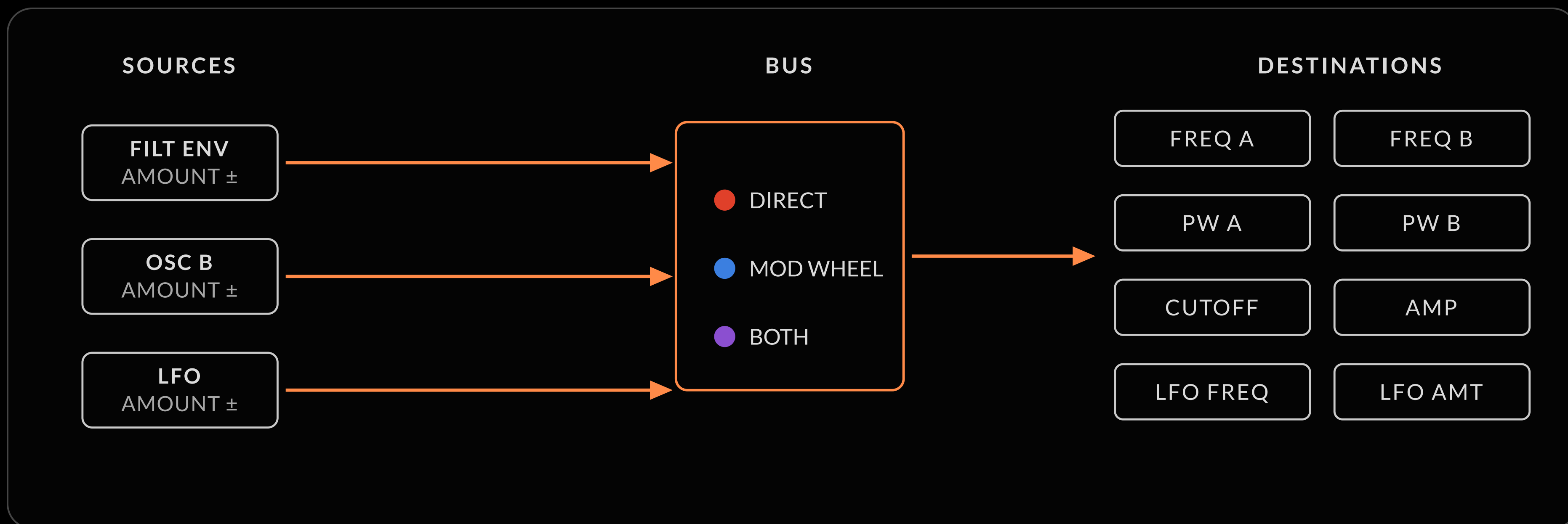
# LOW FREQUENCY OSCILLATOR

Mod wheel up for vibrato. FREQUENCY sets the speed, the shape buttons the motion, AMOUNT how far it goes.



# THE MODULATION SECTION

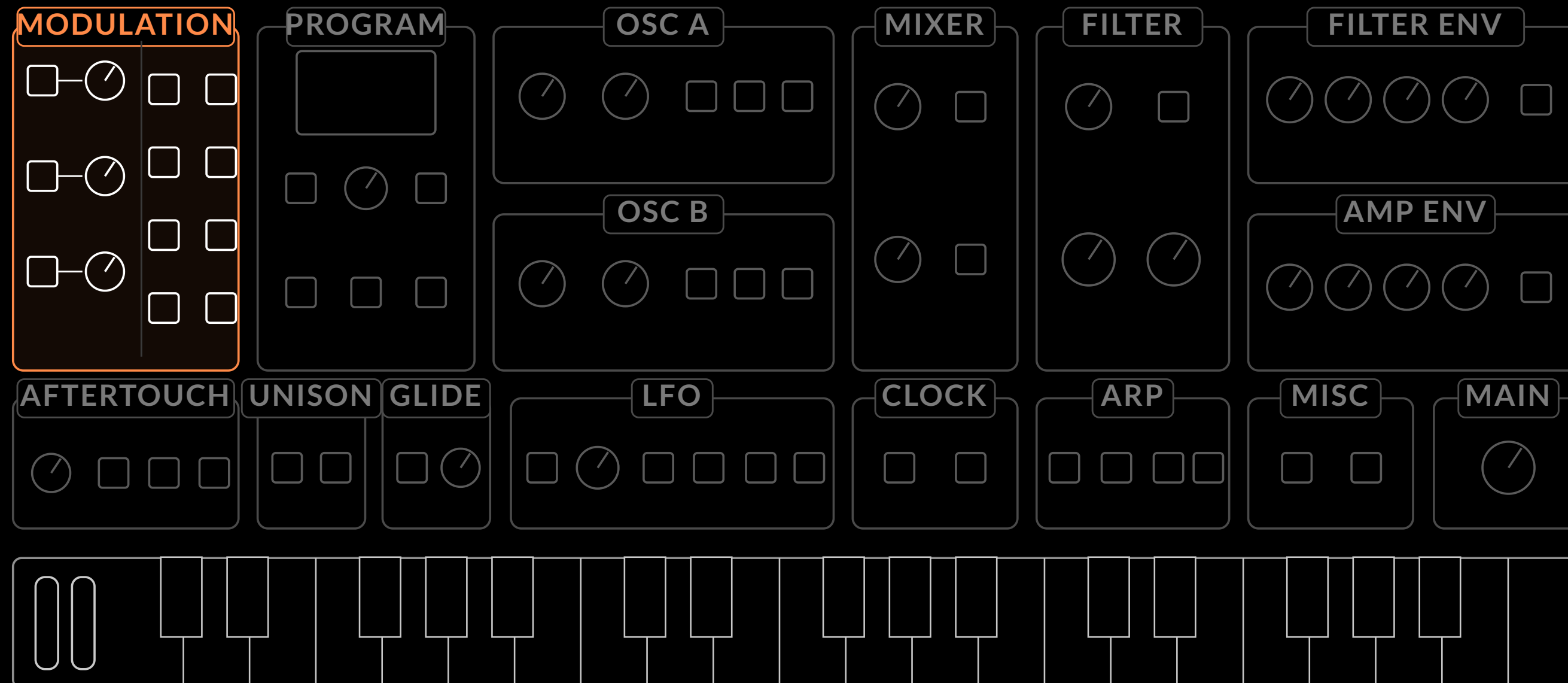
Route → destination → amount. On the wheel bus, the mod wheel is the volume of the modulation.



WHERE ON FOURM

# MODULATION › SOURCE · DESTINATION · AMOUNT

Three slots. Pick a source and a destination, then AMOUNT: red is negative, blue is positive.



# KEY TERMS

New words from this module

## **LFO**

An oscillator too slow to hear.

## **VIBRATO**

LFO to pitch.

## **PWM**

LFO to pulse width.

## **MOD WHEEL**

The volume of the modulation.

## **RATE**

Its speed; SYNC locks it to the clock.

## **TREMOLO**

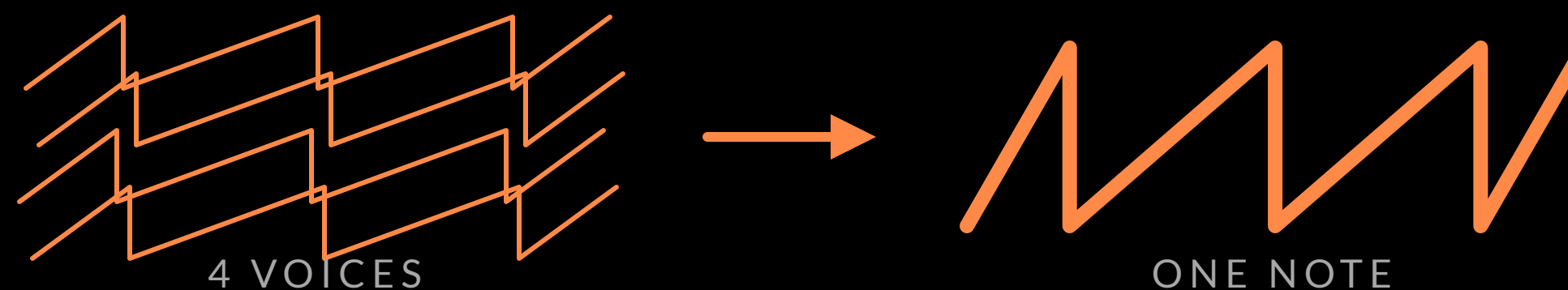
LFO to amplitude.

## **MODULATION BUS**

Direct (red) or mod wheel (blue).

# VOICES & UNISON

Four voices, one note, and the loop that thickens everything

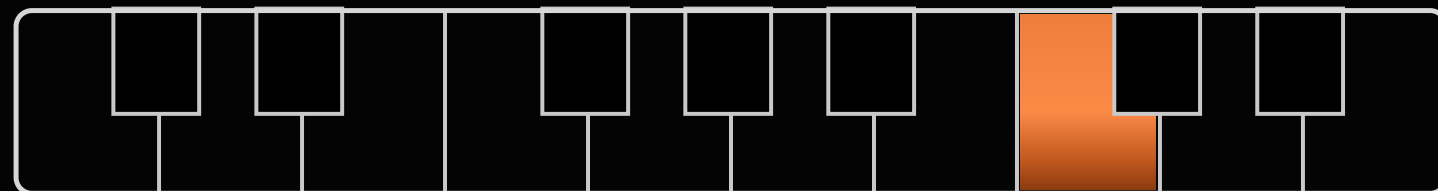


# POLYPHONY

Fourm has four complete voices. Play a fifth note and the oldest one is stolen.

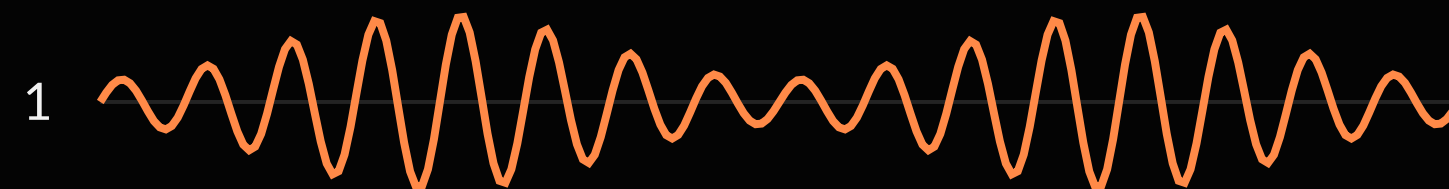
## MONOPHONIC · ONE VOICE

WHAT YOU PLAY



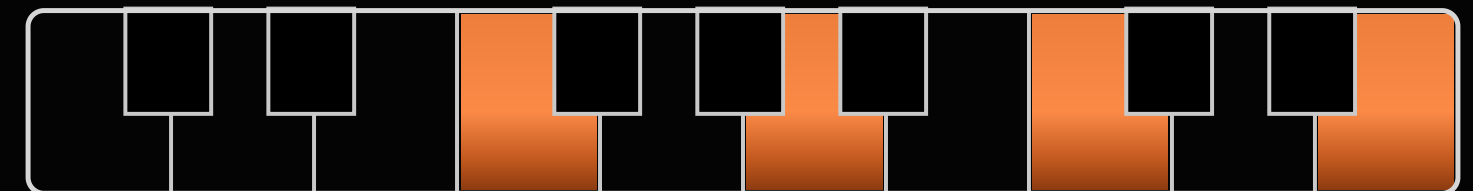
↓ LAST NOTE

WHAT YOU HEAR



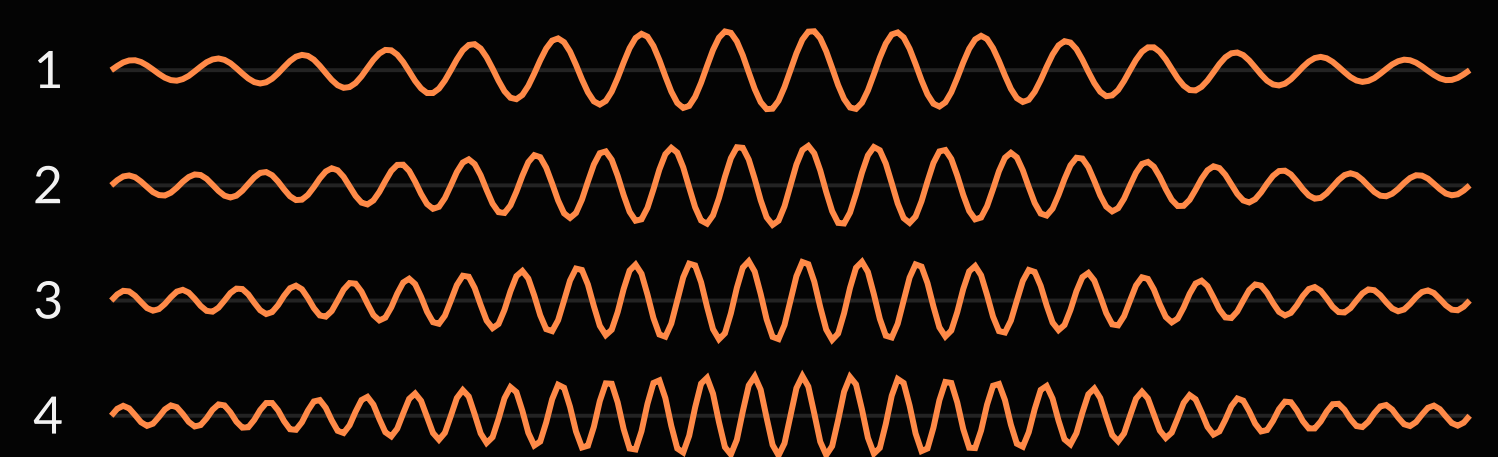
## 4-VOICE POLYPHONIC

WHAT YOU PLAY



↓ ↓ ↓ ↓

WHAT YOU HEAR



# UNISON

Four voices on one note



1-4 VOICES, SLIGHTLY DETUNED



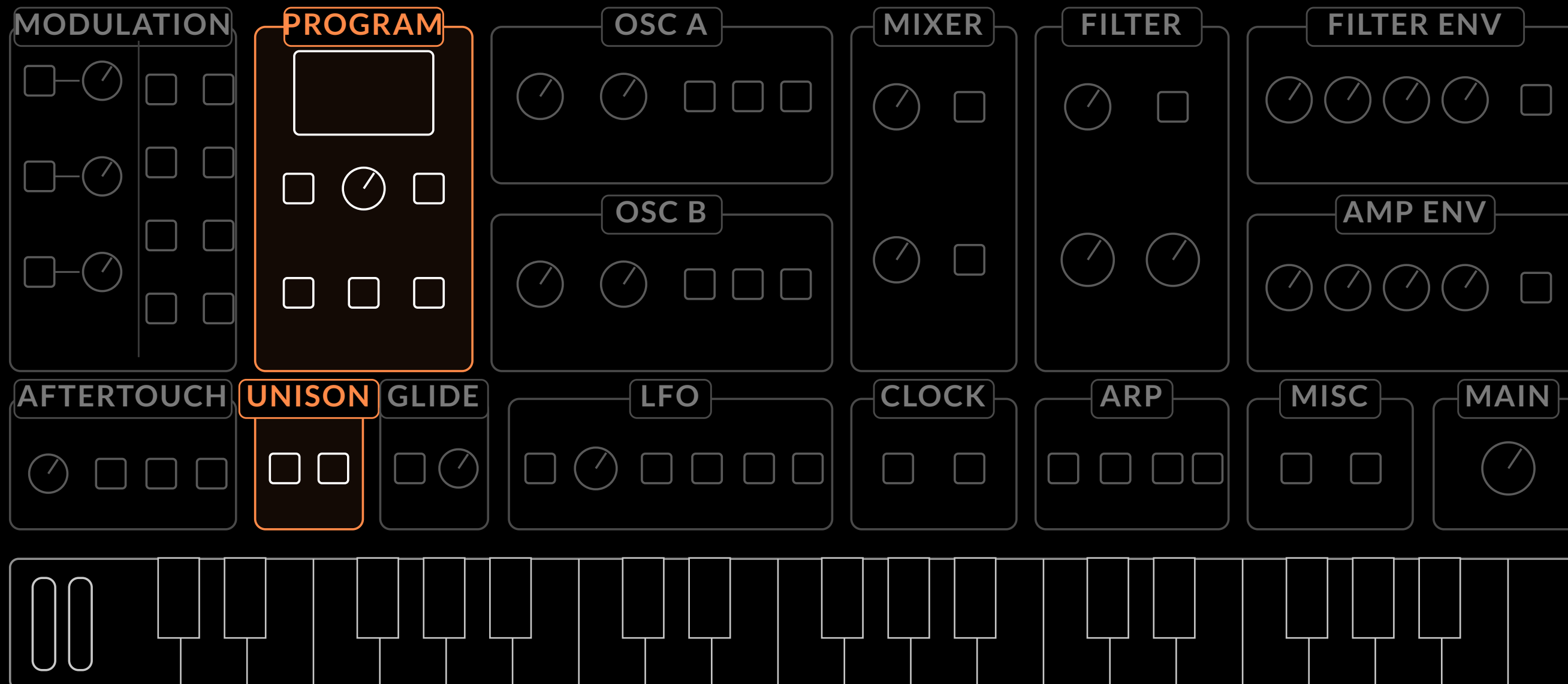
ONE HUGE MONOPHONIC NOTE

**HOLD UNISON + SELECT FOR VOICES • PARAM 10: UNISON DETUNE 0-127**

WHERE ON FOURM

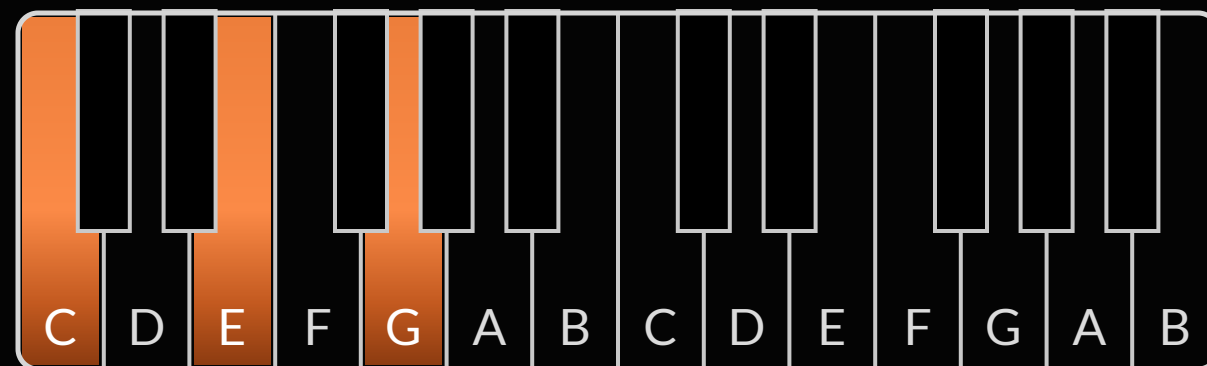
# UNISON + SELECT · DISPLAY: PARAM 10

UNISON on; hold it and turn SELECT for the voice count. The detune between them lives at Param 10 on the display.

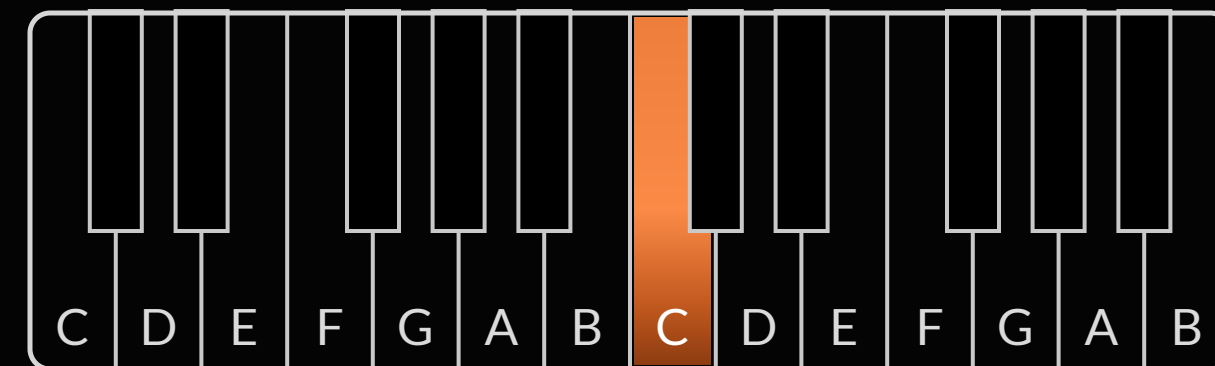


# CHORD MEMORY

One key plays the whole chord



1 · HOLD A CHORD, PRESS UNISON



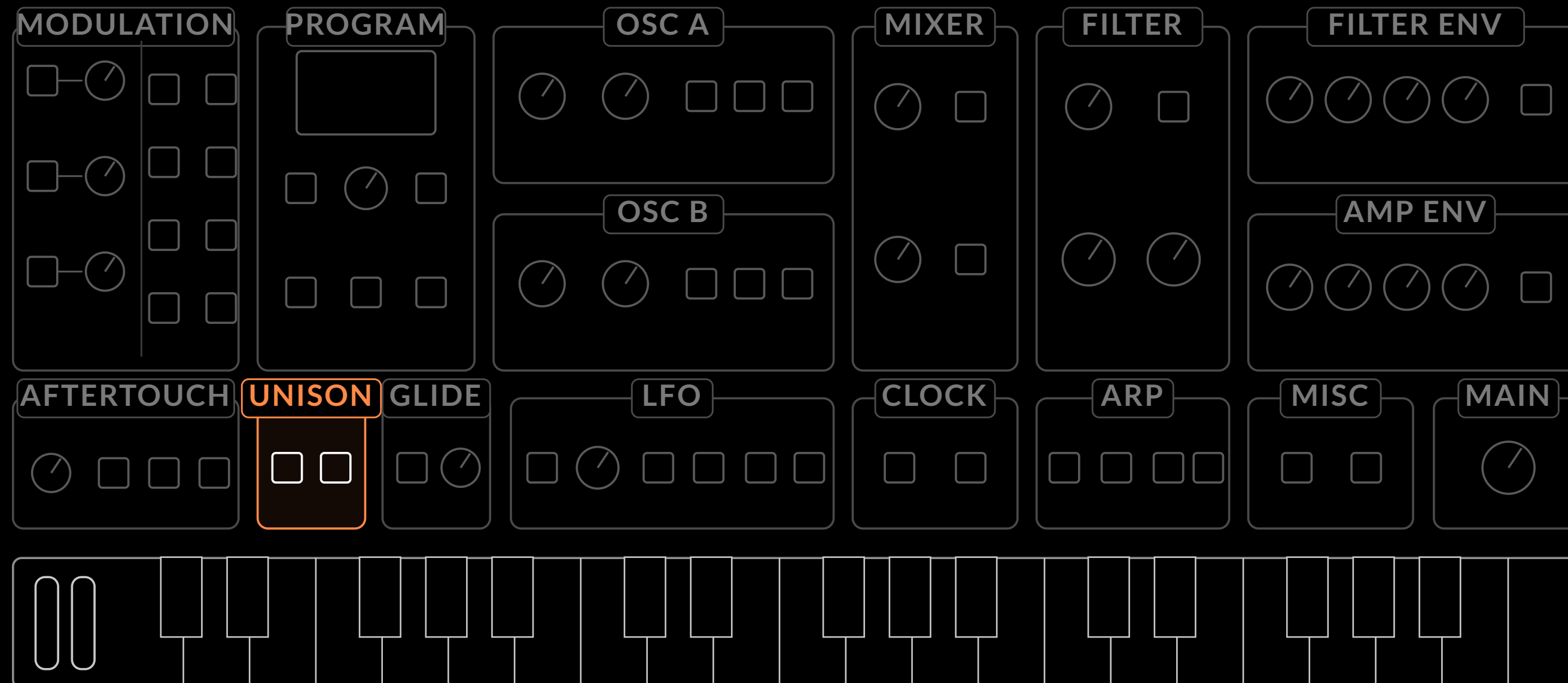
2 · ANY SINGLE KEY REPLAYS IT, TRANSPOSED

**SAVED WITH THE PROGRAM: GREAT FOR CHORD STABS**

WHERE ON FOURM

# UNISON › CHORD MEMORY

Hold a chord and press UNISON: from now on any single key plays the whole chord, transposed.



# FEEDBACK

The output fed back into the mixer, soft-clipped: warmth, then growl.

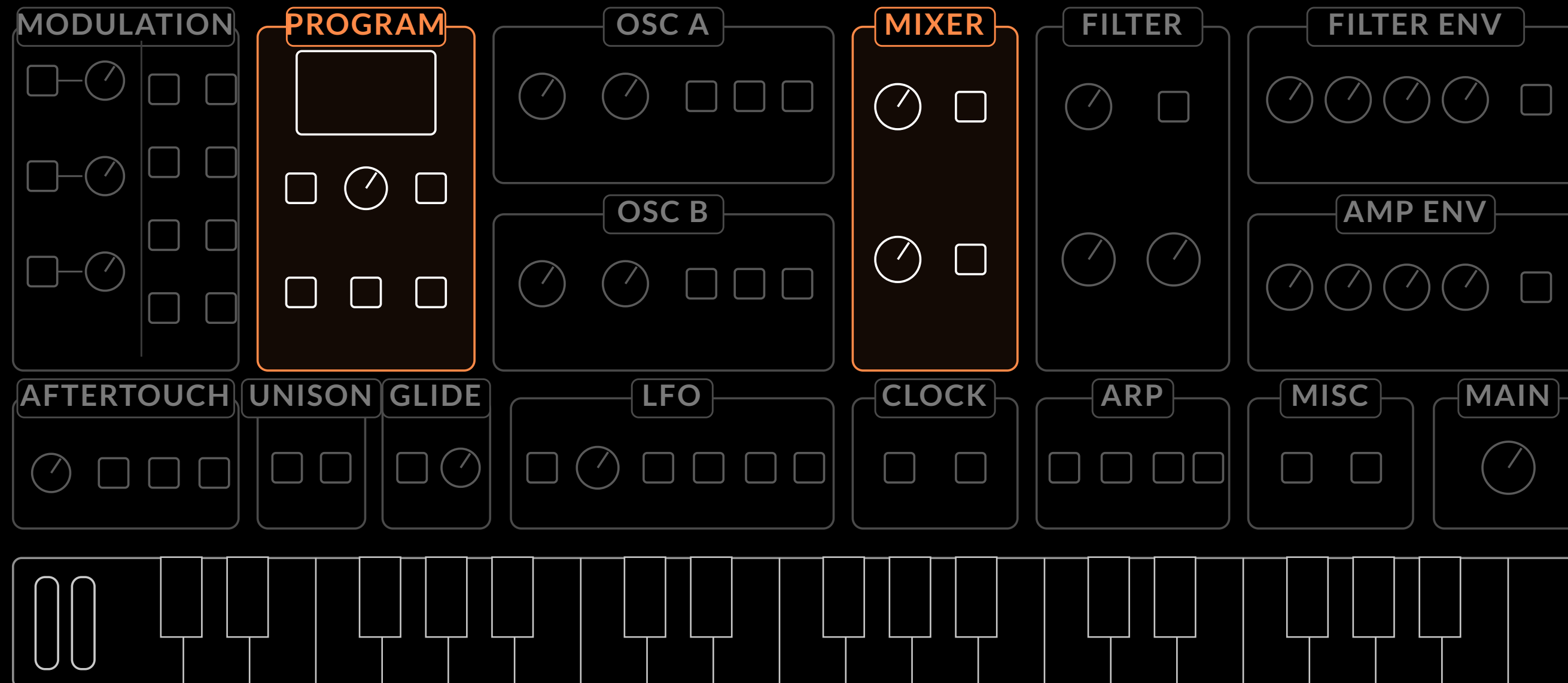


**DEFAULT LEVEL 100 • HOLD FEEDBACK + TURN SELECT • THICKEN, DRIVE, CHAOS**

WHERE ON FOURM

# MIXER › FEEDBACK + SELECT

FEEDBACK on; hold it and turn SELECT for the amount: warmth first, then growl.



# KEY TERMS

New words from this module

## UNISON

All chosen voices play the same note. Monophonic, massive.

## CHORD MEMORY

Unison stores a chord; single keys replay it transposed.

## FEEDBACK

The output fed back into the mixer, soft-clipped.

## POLYPHONY • VOICE

Four complete voices; the fifth note steals the oldest.

## UNISON DETUNE

0–127 of drift between stacked voices: the chorused fat.

## HOLD

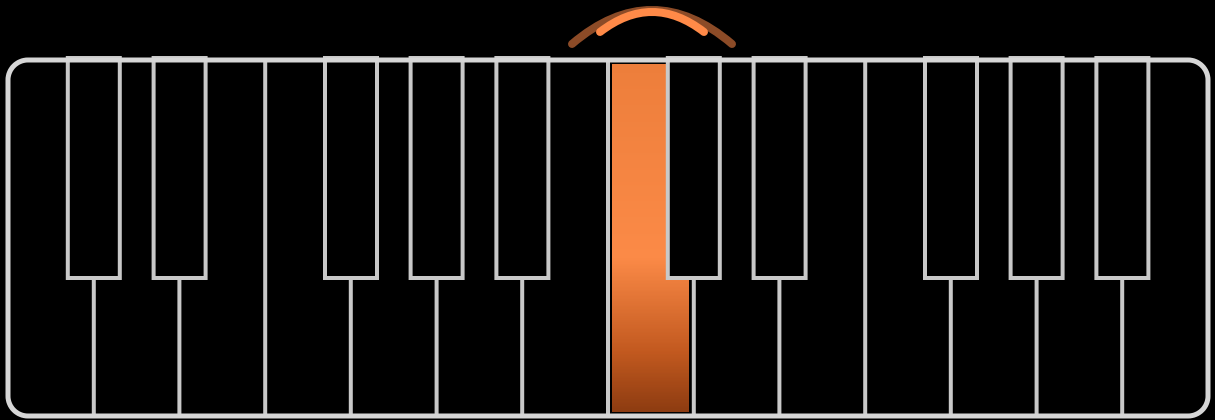
Sustains up to four notes after your hands leave the keys; also latches the arpeggiator.

## SOFT-CLIPPING

Gentle rounding of peaks: saturation instead of harsh distortion.

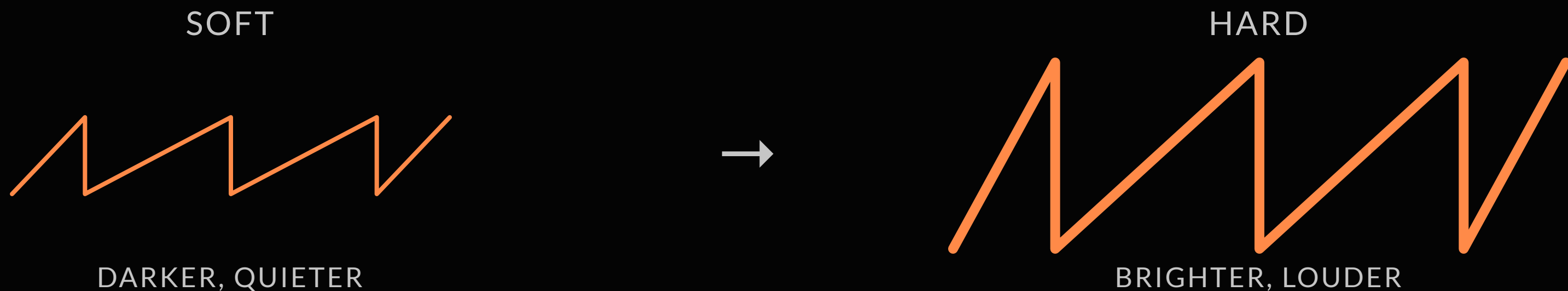
# EXPRESSION

Play it, don't just press it



# VELOCITY

How fast the key goes down. Scales the amp and filter envelopes; five response curves in Global.



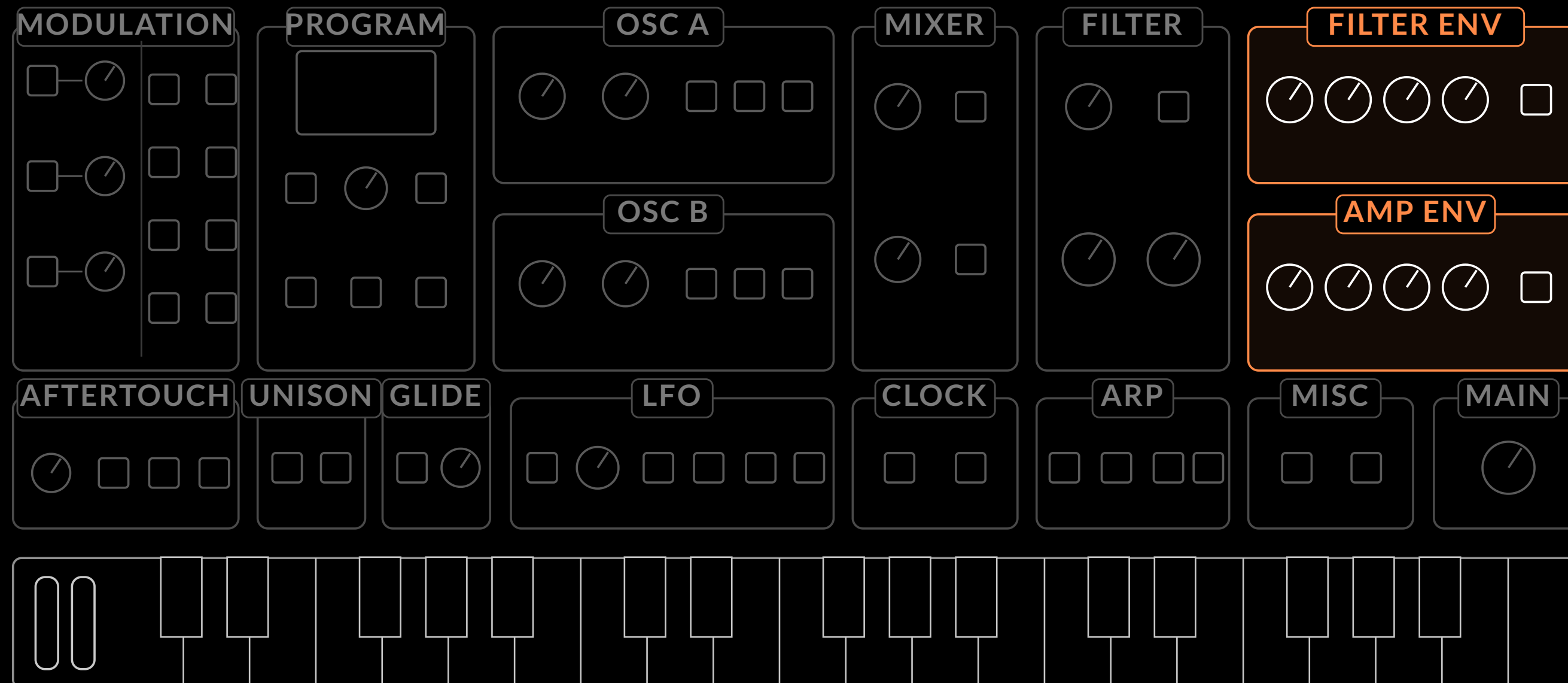
**VELOCITY FEEDS BOTH ENVELOPES: FILT ENV VEL & AMP ENV VEL, 0-127**

FIVE RESPONSE CURVES IN GLOBAL: SOFT+, SOFT, MEDIUM, HARD, HARD+

WHERE ON FOURM

# ENVELOPES › VELOCITY

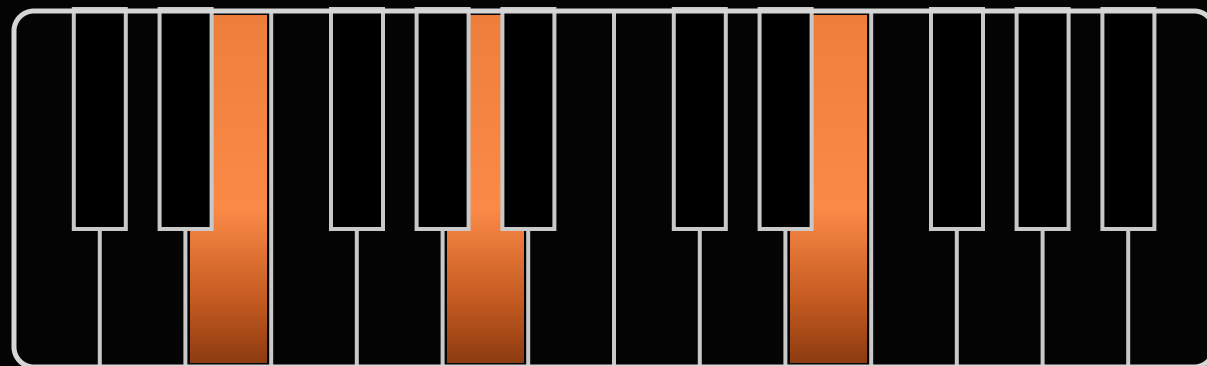
Each envelope has a VELOCITY button: how hard you strike scales that envelope's amount.



# POLY AFTERTOUCH

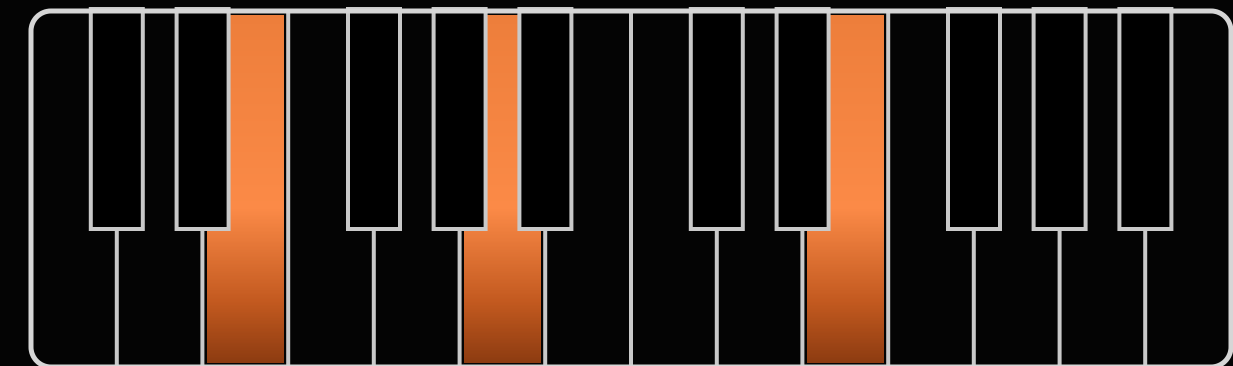
Pressure, per key: the Tactive keybed

MONO AT



ONE PRESSURE VALUE: EVERY VOICE MOVES

POLY AT



A SENSOR UNDER EVERY KEY: ONLY THE PRESSED VOICE MOVES

# PRESSURE ROUTING

Press into one note of a chord and only that note opens.

FREQ A

FREQ B

FILT

AMP

LFO FREQ

LFO AMT

PICK DESTINATIONS WITH THE DEST BUTTONS: TWO PER BUS

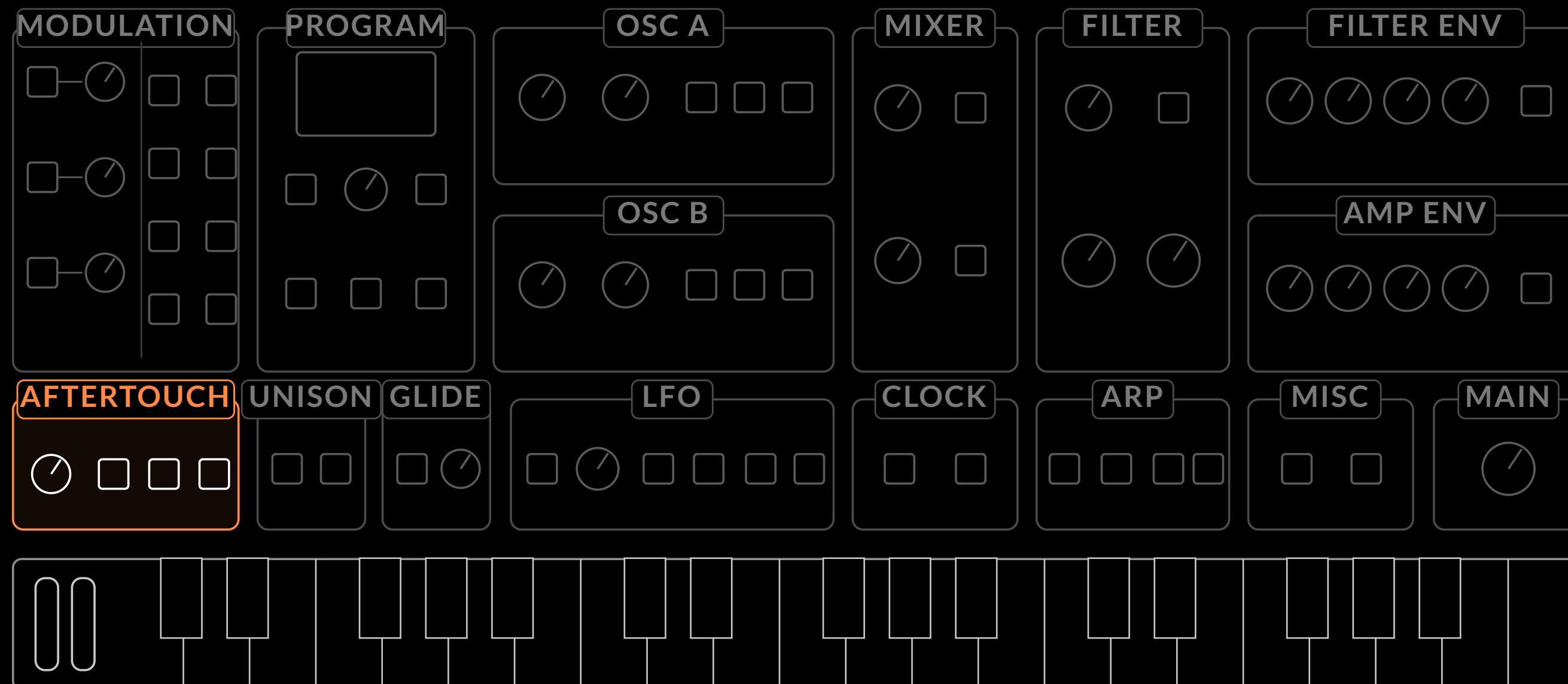
**ONE AMOUNT KNOB, POSITIVE OR NEGATIVE**

PRESS HARDER WITH +FILT: THE SOUND OPENS · WITH -FILT: IT CLOSES

WHERE ON FOURM

# AFTERTOUCH › DESTINATION · AMOUNT

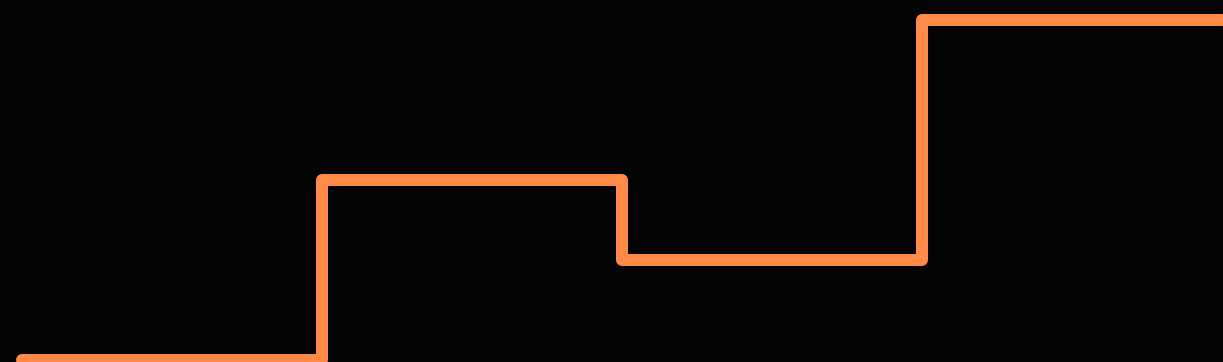
Press harder into a held key. AMOUNT sets how much; the buttons pick where the pressure goes.



# GLIDE

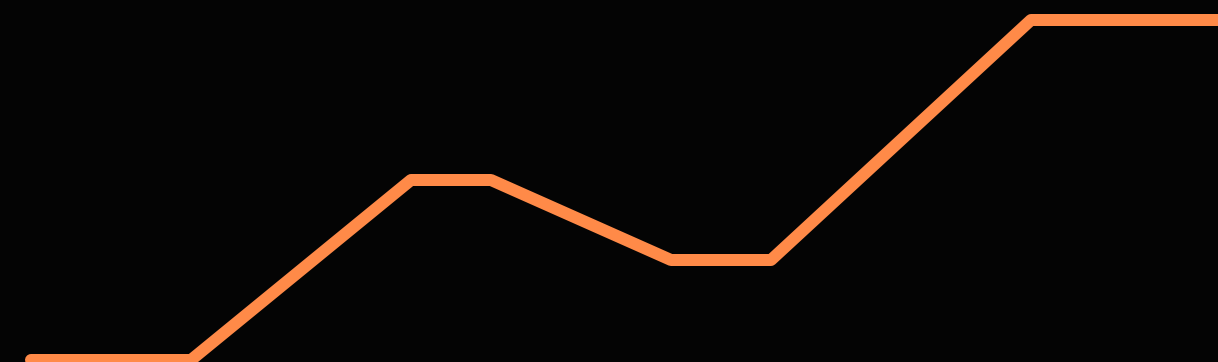
GLIDE on and RATE above zero. Modes: hold GLIDE and turn SELECT.

GLIDE OFF



PITCH JUMPS INSTANTLY

GLIDE ON + RATE



PITCH SLIDES BETWEEN NOTES

**GLIDE ON WITH RATE AT 0 DOES NOTHING. SET BOTH**

# KEY TERMS

New words from this module

## VELOCITY

How fast a key goes down. Routed to filter and amp envelope amounts.

## POLY vs MONO AT

Per-key pressure vs one value for the whole keyboard.

## CURVE

Five velocity curves, from Soft+ to Hard+; aftertouch has its own five.

## GLIDE • PORTAMENTO

Pitch slides between notes instead of jumping; legato modes slide only on overlap.

## AFTERTOUCH (AT)

Pressure applied after the key bottoms out.

## TACTIVE KEYBED

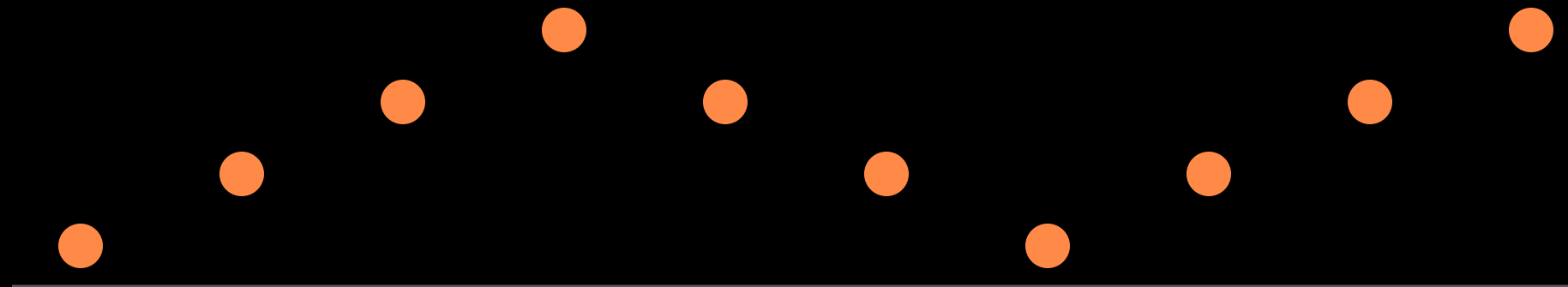
This synth's keybed: a silicone-enclosed pressure sensor per key.

## AMOUNT ±

Modulation depth; negative amounts invert the direction.

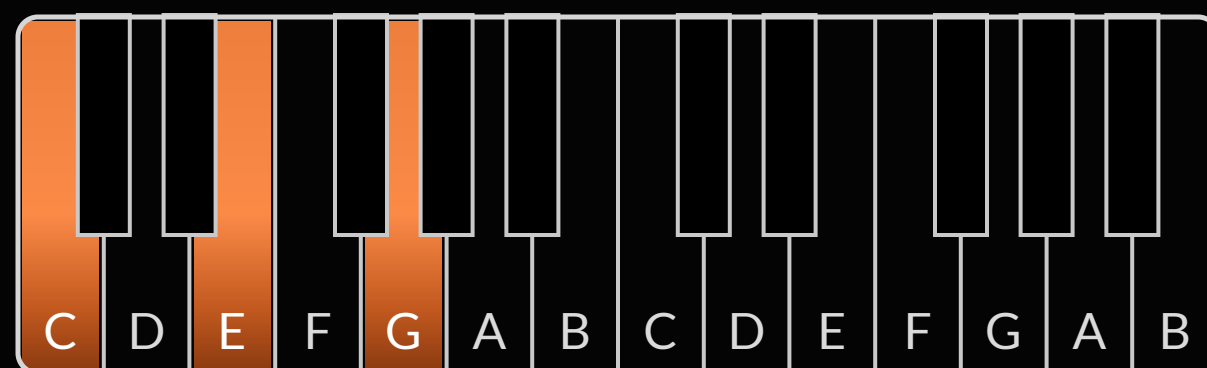
# ARPEGGIATOR & SEQUENCER

Patterns from held notes

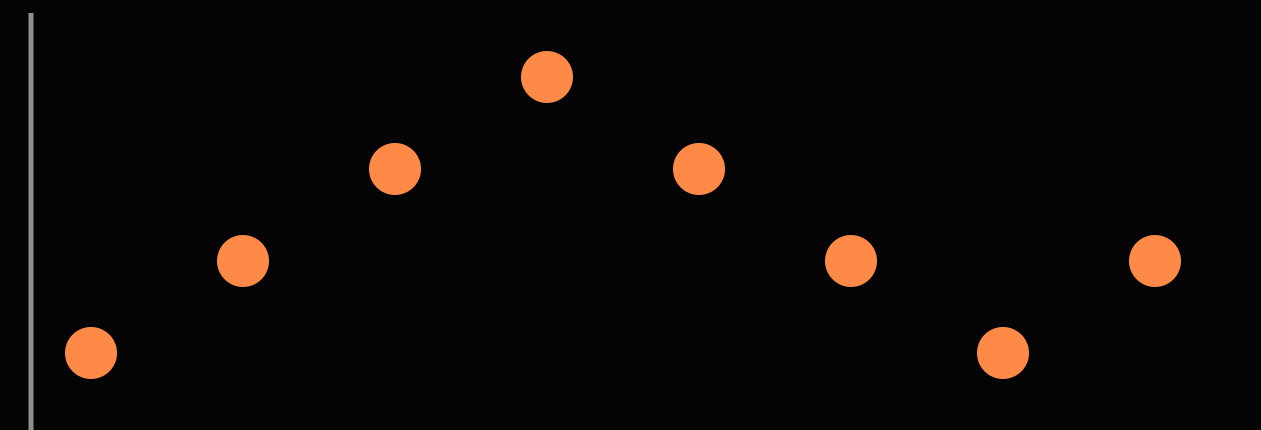


# HOLD A CHORD

Hold a chord and the synth plays the pattern.



HOLD C · E · G



UP+DOWN 1 · OCTAVE 1

**MODES: UP · DOWN · UP+DOWN 1 & 2 · RANDOM · ASSIGN. PRESS HOLD TO LATCH**

WHERE ON FOURM

# ARPEGGIATOR › ON · HOLD + CLOCK

ARP on and hold a chord. HOLD latches it after your hands leave; the CLOCK section sets the tempo.



# CLOCK & SYNC

Tap TAP TEMPO four times, or hold it and turn SELECT. CLK DIV picks the grid.

Clock Mode set to In: the arp waits for external clock.



SET TEMPO: TAP 4× ON TAP TEMPO · OR HOLD BPM + SELECT

CLK DIV

LFO SYNC

MIDI IN/THRU

**WITH MIDI CLOCK MODE IN, THE ARP WAITS FOR EXTERNAL CLOCK**

# NOTE & MOD SEQ

A sequencer in disguise



RECORD UP TO 64 STEPS: NOTES, RESTS, TIES, EVEN GLIDES

SEQ MOD SENDS STEPS TO: OSC FREQ, CUTOFF, PW, FEEDBACK, MOD AMOUNTS

**PLAY ANY KEY: THE WHOLE SEQUENCE TRANSPOSES**

# KEY TERMS

New words from this module

## **ARPEGGIATOR**

Plays a pattern from the notes of a held chord.

## **RELATCH**

Each new chord starts a fresh arpeggio instead of adding notes.

## **CLOCK DIVISION**

Note value per step: quarters, eighths, sixteenths...

## **MOD SEQUENCE**

Steps that move a parameter instead of playing notes.

## **HOLD / LATCH**

Keeps the arpeggio running after you release the keys.

## **BPM / TAP TEMPO**

The clock speed: dialed in or tapped in.

## **REST • TIE**

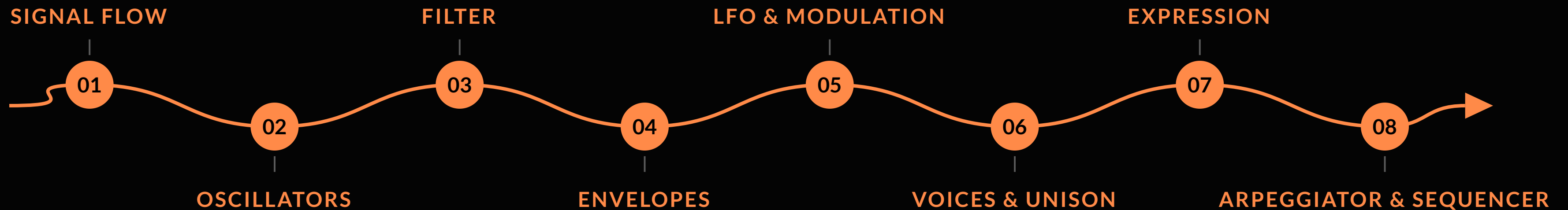
A silent step • a step that extends the previous note.

## **SEQ REVIEW**

Global setting for stepping through and editing a sequence.

# THE ROAD, AGAIN

Every synth is this road with different clothes.





# THANK YOU

Take the handbook home — EN or RO

## SINTEZAUR

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# GLOSSARY

The full glossary is in your handbook



